according to Santa Fe standards. This row is located in the front wall of the cab in order to apply the cab and brace it satisfactorily and to obtain the proper clearance for the cab door. Six solid radial stays are applied under the dynamo bracket.

Standard lengths of staybolts are used throughout, the longest being 12¾ inches over the threads. All staybolts have a 3/16-inch hole drilled 1¼ inches deep in the upper end . After the solid staybolts have been driven, teltale holes are countersunk 1/16-inch deep with a drill having the cutting edges at a 60-degree angle with the center line of the drill. The pitch of staybolts and radial stays center to center in the firebox sheet does not exceed 4 inches in any case. Common staybolts are screwed and riveted to the inside and outside sheets and Tate flexible stays and adjustable crown stays are screwed and riveted to the inside sheet according to the railroad company's practice. Radial stays and staybolts are snapped on the inside end as well as on the outside. Rigid and hollow water space staybolts and Tate flexible staybolts have diameters of 15/16 inch. Radial stays in the crown sheet have a tapered thread 1 3/16 inches in diameter at dimension *B* noted on the detail drawing of the stay. The last row of radial stays is located adjacent to the door sheet in order to thoroughly stay the back portion of the crown.

Staybolts behind the brick work have 3/16-inch holes throughout their full length. Such staybolts as would come behind the brick if the engines were later equipped for burning oil also have 3/16-inch holes drilled in them. Where any of the hollow staybolts installed would not take the gage prescribed by the Bureau of Locomotive Inspection, holes were reamed out sufficiently to meet the requirements.

The first row of staybolts around the fire door opening was not driven until the door seam had been welded. All water space stays, including Tate and hollow stays, have Whitworth threads, 12 per inch. Caps and sleeves for Tate water space and crown bolts have V threads and crown bolts have Whitworth threads at both ends, 12 per inch. Cab knees, brackets and other attachments located over staybolts have holes drilled opposite the stays to facilitate inspection and testing.

ASHPANS

The ashpan is arranged to have the upper slope at as great an angle with the horizontal as possible. The slope at the lower portion has a minimum clearance at the frame of ½ inch. The ashpan of each locomotive has one hopper at the rear and two hoppers in front of the rear truck axle. The ashpan air space around the mud ring is 6 inches and the slope of the tube sheet commences at the inside edge of the spacer support. The flare of the pan extends at least 8½ inches beyond the edge of the mud ring in order to obtain a full air opening. The edge of the flare is turned up for a height of 2 inches. Hinged to this flare a netting covers the air opening at the sides. The pan is built wide enough so that when the netting is applied it does not limit the access of air into the pan. The air opening is about 15 percent of the grate area. The netting is of rectangular form to give the greatest area of opening.

GRATES

The grate fingers, bars and the like are of Santa Fe standard design for bituminous coal but modified in this case to suit the size of the firebox. The top line of the grate is 4¾ inches above the bottom of the mud ring. The side and center grate bar supports are so designed and fastened as not to be moved by operation of the grate shaker. After installation the grates were operated in test to see that the supports were stationary and that the fingers on the bars did not interfere.

The grate rods are attached to the lug on the grate connecting bar. Auxiliary shafts are applied between the back of the ashpan and rear furnace bearer sheet to provide for the application of the shaker rods to the arms on the grate bars near the center of the engine. The dump grates are 17¼ inches wide and are specially reinforced. The grate shaker locks outside and inside are of cast steel designed according to Santa Fe standards instead of the type formerly used in locomotives of this type in order to provide an arrangement that will give satisfactory movement of the grates for road shaking by throwing out the inside locks only. The grate shaker arrangement was designed so that the grates could be operated by either hand or power. The shaker lever latch is above the firing deck in order that it may be operated by foot.

TABLE 1.—PRINCIPAL DETAILS OF 2-10-2 SANTA FE LOCOMOTIVE BOILERS

Style	Conical
Working pressure	195 pounds
Diameter first shell, outside	88 inches
Diameter rear shell, outside	100 inches
Length firebox	132 5/16 inches
Width firebox	96 inches
Thickness firebox sheet	⅜ inch
Thickness first course of shell	13/16 inch
Thickness second course of shell	29/32 inch
Thickness third course of shell	15/16 inch
Wrapper sheet is in three pieces.	
Thickness wrapper sheet top	9/16 inch
Thickness wrapper sheet sides	7/16 inch
Thickness boiler back head	½ inch
Thickness firebox extension wrapper sheet	⅜ inch
Thickness firebox extension throat sheet	½ inch
Thickness outside throat sheet	15/16 inch
Thickness front flue sheet	9/16 inch
Thickness smokebox	⅝ inch
Thickness smokebox liner	¾ inch
Tubes, number	50
Tubes, diameter	5½ inches
Flues, number	275
Flues, diameter	2¼ inches
Arch tubes, number	4
Arch tubes, diameter	3½ inches
Tubes and flues, length	21 feet
Heating surface, tubes	3,388 square feet
Heating surface, flues	1,506 square feet
Heating surface, firebox	376 square feet
Heating surface, arch tubes	41 square feet
Heating surface, total	5,311 square feet
Heating surface, superheater	1,298 square feet
Equivalent heating surface*	7,258 square feet
Grate area	88.3 square feet

TABLE 2.—STAYBOLT INSTALLATION IN 2-10-2 SANTA FE LOCOMOTIVE

Number of crown stays	454
Diameter of crown stays	1 3/16 inches
Number of screw stays	883
Diameter of screw stays	15/16 inch
Number solid radial stays	414
Diameter of solid radial stays	1 3/16 inches
Number Rome hollow stays	306
Diameter Rome hollow stays	15/16 inch
Number Tate flexible stays	971
Diameter Tate flexible stays	15/16 inch
Number Tate flush stays	30
Diameter Tate flush stays	15/16 inch
Number Tate flexible stays	4
Diameter Tate flexible stays	1 ⅛ inches
Number screw stays	28
Diameter screw stays	1 ⅛ inches

The grate shaker arrangement is such that the cylinders are operated independently and located beneath the cab outside the wind sheets at the rear of the boiler back head. One cylinder is located on each side of the locomotive. The cylinder lever jaws, shaker rod jaws and all operating details have the jaws and arms welded instead of bolted together. The shaker cylinders are arranged so that one cylinder operates the right side grates both front and back and the other cylinder the left side grates.

* Equivalent heating surface = total heating surface + 1.5 times the superheating surface.

(*Continued on page* 124)

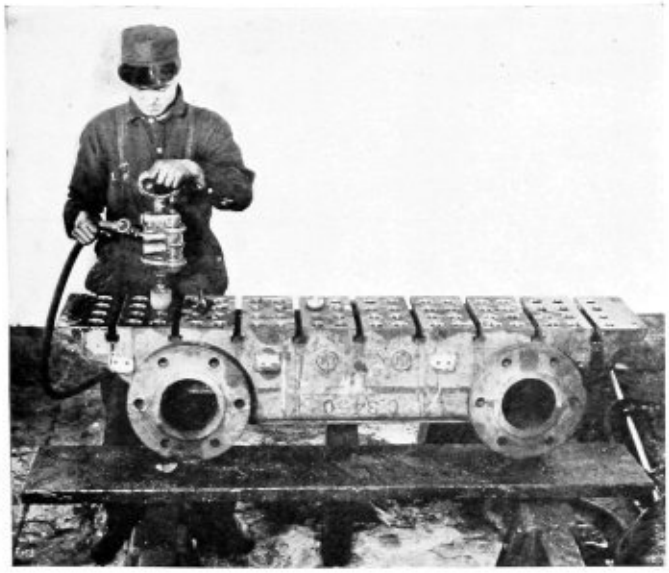


Fig. 7.—Operation of Grinding Unit Seats on Header

Beads should have the fins on the outer edges cut off with a narrow cape chisel when formed. The heel of the beading tool should not be used for cutting off; to do so means that the flue will be loosened in the sheet and eventually the sheet grooved around the edge of the bead.

CLEANING IN ROUNDHOUSE

The flues and the outside surface of the units should be kept clean and all dirt, cinders and soot removed from the flue. The back end of the return bends should have all clinkers or cinders broken off and removed from the firebox end.

In cleaning the flue, water should be used in preference to air. Washing the flues can be done by the boiler washout gang and with the same tools employed for this work. Special care must be taken to see that the flues are thoroughly

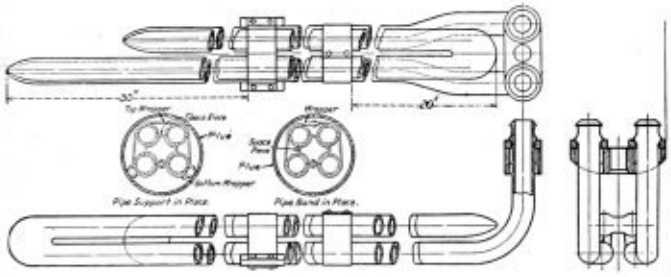


Fig. 8.—Location of Bands and Supports on Return Bend Units

cleaned. If the hole in the end of the pipe is reduced to the nominal diameter of the pipe, i. e., ½-inch hole in ½-inch pipe, etc., better results will be obtained.

If air is used for cleaning the flues, insert the pipe in the flue, under the unit, for its full length, then open the air valve and gradually draw the pipe back through the flue. This procedure avoids clogging the flue, which would result if the air pipe were worked forward. After the first operation, the air pipe can be worked back and forth until the flue is entirely clean. The superheater damper should always be open when blowing out flues.

A protection from the ejected dirt may be secured by the use of a disk of stiff leather or wood about 9 inches in diameter, with a hole in its center, which is a close sliding fit on the cleaning pipe. Place this on the pipe so as to stand about ½ inch away from the flue sheet before turning on the air.

After the flues are clean they should be inspected from the firebox end while a light is held at the front end to insure that the bottom of the flue is clean. The test of the draft drawing the torch flame into the flue is not satisfactory, as the flame will be drawn into the flue by the draft over the top of the dirt while the whole bottom of the flue may be stopped.

In case water instead of air is used to clean the flues, hot water should be used and the boiler should have steam on so that the flues, units, etc., will dry rapidly and so partially avoid the rust and corrosion following such practice.

CUTTING OUT FLUES

In cutting out superheater flues a single rotation cutting-off tool has been developed which gives satisfaction. It is used with the same motor and a speed reducing rig similar to that used on the smaller flues. In cutting off the flue at the front end, the cut should be made as close to the sheet as seems safe. At the firebox end the cut should be far enough ahead in the flue to cut off the prossered part of the flue, so as to facilitate the application of the safe-end at this end of the flue.

If the flue is cut out in the old method of slitting with the cape chisel, the beads at the front end of the flue should be removed before the flue is placed in the rattler, to prevent

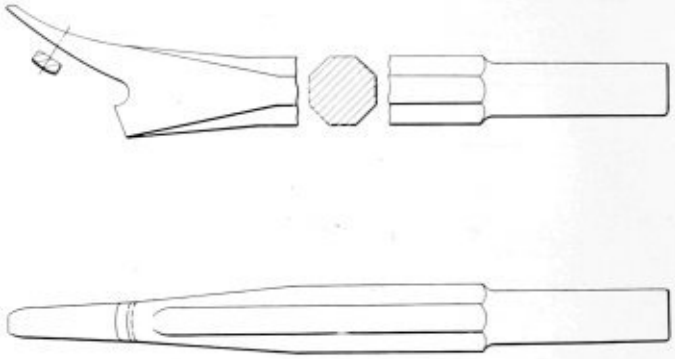


Fig. 9.—Standard Tool for Beading Flues

breaking or cracking of the large end of the flue, which commonly occurs where the beads are not removed.

If the flues should be dented in any manner, straighten out the dents before the flue is replaced in the boiler, to permit the easy insertion of the unit.

SAFE ENDING FLUES

In cutting off the flue, preparatory to safe-ending, the swaged end should be cut off beyond any previous welds that have been made, thereby maintaining only one weld on the swaged end of the flue. When the swaged end has been reduced to its limit by this process, it should be reswaged and sufficient material of standard diameter welded to the large end to provide the necessary length. (This weld should be so made that no obstruction will appear in the inside of the flue.) In applying the safe end to the swaged end of the flue, bell the end of the flue and scarf the safe end about 5/8 inch. In heating for welding watch the inside of the flue to insure the proper heat on the safe end. Better results can be obtained if the safe end is annealed after having been applied.

In applying safe ends to the front end of the flues, the weld should be smooth inside so as not to interfere with insertion of the unit and smooth outside so as not to interfere with application of the flue.

TESTS

The superheater, steam and exhaust pipe joints and dry pipe joint should be tested periodically every thirty days if convenient, at the time of the regular washout, with at least